Effect of Seed Priming on Germination and Seedling Development of Pepper under Salinity Stress



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Abstract

Salinity stress affects crop production in various stages of plant growth with germination and seed development being the most sensitive to this stress. The need to raise uniform seedlings with better crop establishment is of utmost importance. Various seed priming methods have been employed in many crops to alleviate the effect of saline, drought and temperature stress. The objective of this study was to evaluate NaCl and PEG priming for enhanced germination and improved seedling establishment in pepper under salinity stress. A study to select a priming protocol using three concentrations (25, 50 and 100 mM) of each priming agent were applied on 'Legon 18' pepper seeds at three different durations of 12, 24 and 36 hours. Results indicated that 25 mM NaCl

applied to ‘Legon 18’ seeds for 36 hours and 25 mM PEG for 24 hours, had the highest germination percentages, germination index, and rate of germination with lowered mean germination time after seeds were sown in seed trays. Further studies using best primed treatments of NaCl and PEG were used to assess their response to germination under salinity stress. Primed and unprimed seeds were sown in transparent plastic cups at different saline levels (0, 4.46 and 8.95 dS/m). The results revealed that NaCl and PEG priming improved germination percentage, germination index, seedling length, mean germination time and seedling vigour than control though there were no significant interaction between salinity and priming among the studied traits. Primed and unprimed seeds were transplanted after 30 days of being irrigated with tap water. Saline treatments (0, 4.46 and 8.95 dS/m) were imposed after transplanting for another 30 days. Priming treatments improved plant height, number of leaves, chlorophyll content, stem diameter and total leaf area although they were reduced by increase salinity. Also, priming treatments generally increased K+/Na+ and Ca+2/Na+ ratios in leaves, stems and roots of pepper seedling plants than in control at all saline levels. From the results, seed priming with NaCl and PEG can be employed to increase germination and seedling performance of Legon 18 pepper under salinity stress.

Description

MPhil.

Keywords

Seed, Pepper, Salinity Stress, Crop production

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